

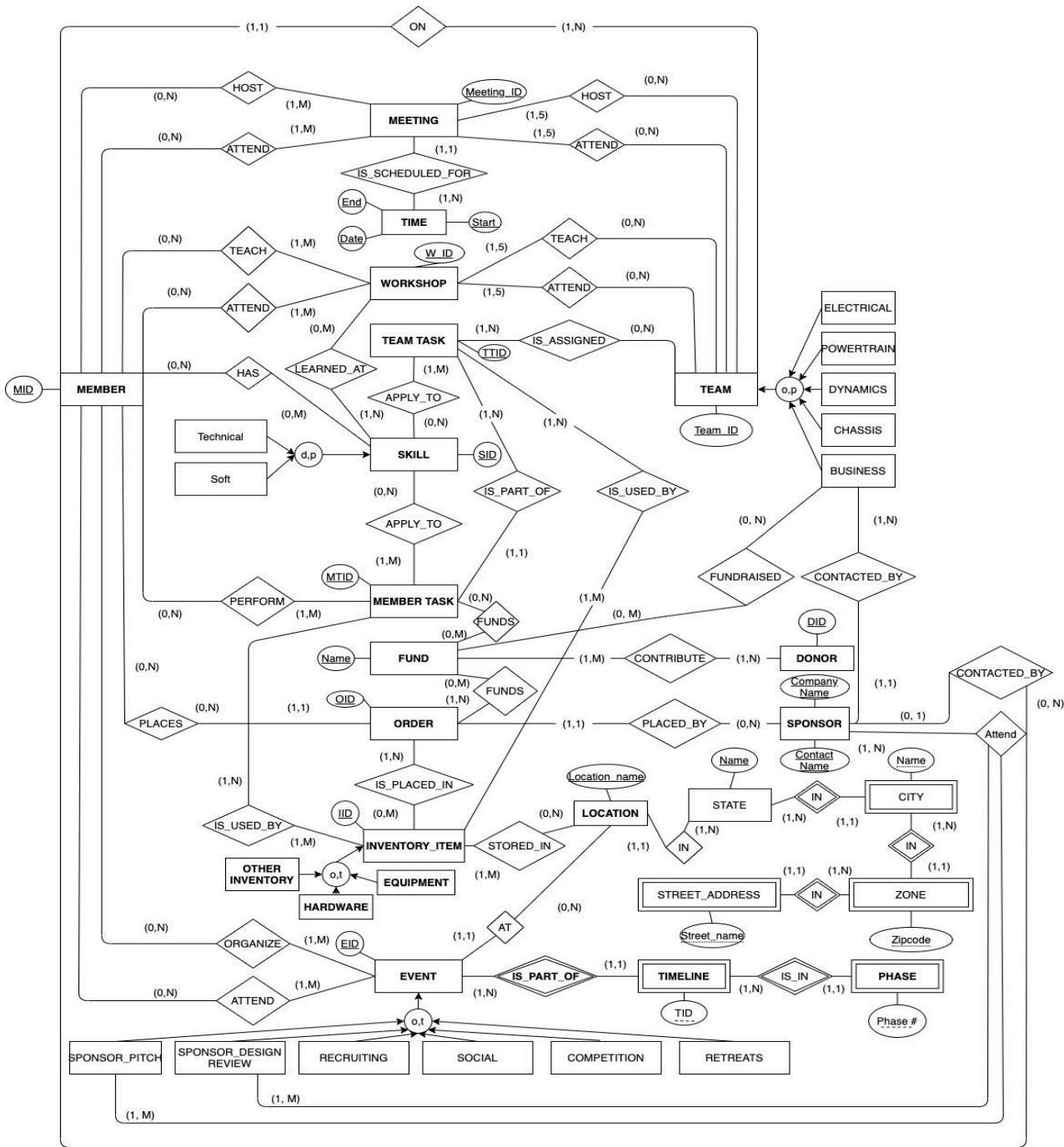
DP Title: DP Review 3 - Formula Electric at Berkeley (FEB) Database

Team Number: 10

Team Members: Kevin Feng, Yanling Liu, Parker Nelson, Rachel Lim,
Carol Lee, Howard Yan, Arsh Vishen, Alisha Laroia, Alan Hu

Client Description: Formula Electric at Berkeley (FEB) is UC Berkeley’s Formula SAE Electric race car team. The team consists of about 40 UC Berkeley students, who are working to design, build, and race Berkeley’s first electric formula-style race car in the Formula SAE EV competition. The contact person is Brad Ling (bradling@berkeley.edu, (909)-348-2331), club president of Formula Electric at Berkeley.

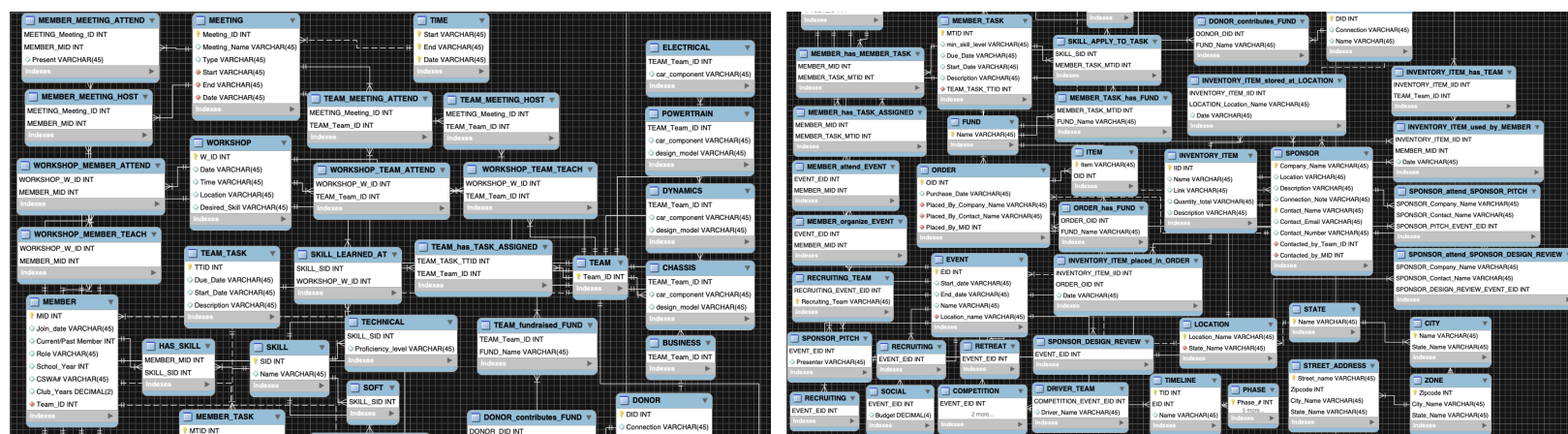
Simplified EER



Relational Schema

1. TEAM_TASK (TTID, Due_Date, Start_Date, Description)
 2. DONOR (DID, Connection, Name)
 3. MEMBER_TASK (MTID, min_skill_level, Due_Date, Start_Date, Description, TTID¹)
 4. INVENTORY_ITEM (IID, Name, Link, Quantity_total, Description)
 - 4a. Other_Inventory (IID⁴, manufacturer)
 - 4b. Hardware (IID⁴, brand)
 - 4c. Equipment (IID⁴, brand)
 5. WORKSHOP (W_ID, Date, Time, Location, Desired_Skill)
 6. LOCATION (Location_Name, State_Name²⁰)
 7. ORDER (OID, Purchase_Date, Placed_By_MID¹³, Placed_By_Company_Name¹², Placed_By_Contact_Name¹²)
 8. EVENT (EID, Optional, Start_date, End_date, Name, Location_Name⁶)
 - 8a. SPONSOR_PITCH (EID⁸, presenter)
 - 8b. SPONSOR_DESIGN_REVIEW (EID⁸)
 - 8c. RECRUITING (EID⁸)
 - 8d. SOCIAL (EID⁸, budget)
 - 8e. COMPETITION (EID⁸, speed_test_result, reliability_test_result)
 - 8f. RETREAT (EID⁸, budget)
 9. SKILL (SID, Name)
 - 9a. TECHNICAL (SID⁹, Proficiency_level)
 - 9b. SOFT (SID⁹)
 10. MEETING (Meeting_ID, Meeting_Name, Type, Start¹¹, End¹¹, Date¹¹)
 11. TIME (Start, End, Date)
 12. SPONSOR (Company_Name, Location, Description, Connection_Note, Contact_Person_Name, Contact_Email, Contact_Number, Contacted_by_Team_ID^{14c}, Contacted_by_MID¹³)
 13. MEMBER (MID, Join_date, Current/Past_Member, Role, School_Year, CSWA#, Club_Years, Team_ID¹⁴)
 14. TEAM (Team_ID)
 - 14a. ELECTRICAL (Team_ID¹⁴, car_component)
 - 14b. POWERTRAIN (Team_ID¹⁴, car_component, design_model)
 - 14c. DYNAMICS (Team_ID¹⁴, car_component, design_model)
 - 14d. CHASSIS (Team_ID¹⁴, car_component, design_model)
 - 14e. BUSINESS (Team_ID¹⁴)
 15. FUND (Name)
 16. ITEM (OID⁷, Item)
 17. RECRUITING_TEAM (EID^{8c}, Recruiting_team)
 18. DRIVER_NAME (EID^{8c}, Driver_name)
 19. STATE (Name)
 20. CITY (Name, State_name¹⁹)
 21. ZONE (Zipcode, City_Name²⁰, State_name¹⁹)
 22. STREET_ADDRESS (Street_Name, Zipcode²¹, City_Name²⁰, State_name¹⁹)
 23. TIMELINE (TID, EID⁸, Name)
 24. PHASE (Phase#, TID²³, EID⁸, Start_date, End_date, Completed_or_not)
 25. TEAM_MEETING_ATTEND (Meeting_ID¹⁰, Team_ID¹⁴)
 26. MEMBER_MEETING_ATTEND (Meeting_ID¹⁰, MID¹³, Present)
 27. TEAM_MEETING_HOST (Meeting_ID¹⁰, Team_ID¹⁴)
 28. MEMBER_MEETING_HOST (Meeting_ID¹⁰, MID¹³)
 29. WORKSHOP_MEMBER_TEACH (MID¹⁰, W_ID⁵)
 30. WORKSHOP_MEMBER_ATTEND (MID¹⁰, W_ID⁵)
 31. WORKSHOP_TEAM_TEACH (Team_ID¹⁴, W_ID⁵)
 32. WORKSHOP_TEAM_ATTEND (Team_ID¹⁴, W_ID⁵)
 33. HAS_SKILL (MID¹⁰, SID⁹)
 34. SKILL_APPLY_TO_TASK (MTID³, SID⁹)
 35. SKILL_LEARNED_AT (SID⁹, W_ID⁵)
 36. MEMBER_has_MEMBER_TASK (MID¹⁰, MTID³)
 37. TASK_ASSIGN (MID¹⁰, MTID³)
 38. TEAM_has_TASK_ASSIGNED (TTID¹, Team_ID¹⁴)
 39. EVENT_ATTEND (EID⁸, MID¹⁰)
 40. MEMBER_organize_EVENT (EID⁸, MID¹⁰)
 41. SPONSOR_attend_SPONSOR_PITCH (Sponsor_Company_Name¹², EID^{8a})
 42. SPONSOR_attend_SPONSOR_DESIGN_REVIEW (Sponsor_Company_Name¹², EID^{8b})
 43. MEMBER_TASK_has_FUND (MTID³, Name¹⁵)
 44. ORDER_has_FUND (OID⁷, Name¹⁵)
 45. TEAM_fundraised_FUND (Team_ID^{14c}, Fund_Name¹⁵)
 46. DONOR_contributes_FUND (DID², Fund_Name¹⁵)
 47. INVENTORY_ITEM_used_by_MEMBER (IID⁴, MTID³, date)
 48. INVENTORY_ITEM_placed_in_ORDER (OID⁷, IID⁴, date)
 49. INVENTORY_ITEM_used_by_TEAM (IID⁴, TTID¹, date)
 50. INVENTORY_ITEM_stored_at_LOCATION (IID⁴, Location_Name⁶)

MySQL Relationship View



Queries

Query 1

Our team is interested in finding the count of members per team and **visualizing this data as a time series** in order to analyze historical growth trends per team and use these insights to inform recruitment decisions. For example, recruitment heads may use the visualizations to note that membership of the business division has been falling over time (i.e. over the past few semesters). **They can use this insight to invest in recruitment into this division.** The required data for this analysis is the count of members in each team over time. We will run the below query for each team, and then generate the corresponding time series plots via python according to the generated data. On the resulting plots (one plot for each team), the x axis will be the join date, and the y axis will be the count of members in that team.

SQL: SELECT join_date, COUNT(*) FROM MEMBER WHERE team_name = "BUSINESS" GROUP BY join_date;

Query 2

Furthermore, our team is working on creating an order forecasting system that tracks the demand of Formula Electric (FE) materials and supplies. FE would like to know what needs to be ordered at the optimal time to meet the predicted demand. Our team will be using historical order and inventory data to create a SARIMA time series and Facebook Prophet model that determines the seasonal demand for certain items. Using the dates and historical shipping times, we will determine the dates that supplies will be used up and produce suggested order dates for new supplies accordingly. This query enables FE to minimize supply and material holding costs. This will enable optimal inventory management, minimal costs, and minimal probability of stockout.

Query 3

This query aims to predict which members should be promoted to leadership positions based on four attributes: participation, leadership, public relations, and skill set. We begin by computing all attributes (as listed below) for past members. Training data will be historical member profiles, where members are labeled 0 if they have not held a leadership position, and 1 otherwise. We will train a logistic regression model that uses computed attributes to predict whether a member has been a leader or not. We will use the model to predict potential leadership outcomes for future candidates. This query would improve FE's evaluation of team member performance. Based on the weighted scores for all computed attributes, the club will be able to measure participation and highlight the most active members who should be considered for future leadership roles.

Sample SQL Queries: Attributes that Measure Participation

Attribute	Explanation	Code
#_WORKSHOP_ATTEND	The number of workshops each member attended.	SELECT MEMBER_MID, COUNT(WORKSHOP_W_ID) AS "NUM_WORKSHOP_ATTENDED" FROM WORKSHOP_MEMBER_ATTEND GROUP BY MEMBER_MID;
#_TASK_PERFORMED	The number of tasks each member has performed.	SELECT MEMBER_MID, COUNT(MEMBER_TASK_MID) AS TASK_PERFORMED FROM MEMBER_has_MEMBER_TASK GROUP BY MEMBER_MID;
MEETING_ATTENDANCE	The percentage of meetings a member was present at.	SELECT m.MID, (a.Mts_Attended / a.Total_Mts) AS MEETING_ATTENDANCE FROM 'MEMBER' m LEFT JOIN (SELECT MEMBER_MID, COUNT(MEETING_Meeting_ID) AS Total_Mts, SUM(Present) as Mts_Attended FROM MEMBER_MEETING_ATTEND GROUP BY MEMBER_MID) AS a ON m.MID = a.MEMBER_MID;
EVENT_ATTENDANCE	The percentage of events each member attended since they joined	SELECT a.MID, (a.EVENT_ATTENDED / b.NUM_EVENT) AS EVENT_ATTENDANCE FROM (SELECT m.MID, COUNT(ea.EVENT_EID) as EVENT_ATTENDED FROM 'MEMBER' m LEFT JOIN MEMBER_attend_EVENT ea ON m.MID=ea.MEMBER_MID GROUP BY m.MID) AS a

	the club.	JOIN (SELECT m.MID, COUNT(e.EID) as NUM_EVENT FROM 'MEMBER' m LEFT JOIN EVENT e ON e.Start_date >= m.Join_date GROUP BY m.MID) AS b ON a.MID=b.MID;
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Sample SQL Queries: Attributes that Measure Leadership, Public Relations, and Skill Set

Category	Attribute	Explanation	Code
Leadership	#_WORKSHOP_TAUGHT & #_EVENT_ORGANIZED	The number of workshops a member has taught and the number of events a member has organized.	select a.MEMBER_MID, COUNT(a.WORKSHOP_W_ID) as WORKSHOP_TAUGHT, COUNT(a.EVENT_EID) as EVENT_ORGANIZED FROM (SELECT wt.MEMBER_MID, wt.WORKSHOP_W_ID, eo.EVENT_EID From WORKSHOP_MEMBER_TEACH as wt LEFT JOIN MEMBER_organize_EVENT as eo On wt.MEMBER_MID=eo.MEMBER_MID UNION ALL select eo.MEMBER_MID, wt.WORKSHOP_W_ID, eo.EVENT_EID From WORKSHOP_MEMBER_TEACH as wt RIGHT JOIN MEMBER_organize_EVENT as eo On wt.MEMBER_MID=eo.MEMBER_MID where wt.MEMBER_MID is null) as a group by a.MEMBER_MID;
Public Relations	#_SPONSOR_ORDERS_OBTAINED	The number of sponsored orders per company a member helped get.	SELECT m.MID, s.Company_Name, COUNT(o.oid) AS No_Sponsor_Orders_Obtained FROM SPONSOR as s INNER JOIN 'MEMBER' as m ON s.Contacted_by_MID = m.MID INNER JOIN 'ORDER' as o ON s.Company_Name = o.Placed_By_Company_Name GROUP BY m.MID, s.Company_Name ORDER BY COUNT(o.oid);
Skill Set	#_SKILLS_LEARNED	The number of skills each member has learned from workshops taught by the club.	SELECT hs.MEMBER_MID, COUNT(hs.SKILL_SID) AS SKILLS_LEARNED FROM HAS_SKILL as hs INNER JOIN SKILL_LEARNED_AT as learn ON hs.SKILL_SID = learn.SKILL_SID GROUP BY hs.MEMBER_MID ORDER BY COUNT(hs.SKILL_SID) DESC;

Queries 4 & 5

	Query 4	Query 5
Descriptions	This query evaluates how engaged existing donors and sponsors are by analyzing their financial contributions. We will look at the materials sponsors have contributed and tailor requests based on the inventory needs for the club. Through this query, we can understand the extent of contributions from existing donors and sponsors. Furthermore, we can send gifts to active donors and sponsors to reinforce club connections.	We design a query to determine most enthusiastic donors by industry and project future donations. We achieve this by grouping donors by industry and assessing their monetary donations. We will also guide sponsor acquisition by determining which sponsors by industry held the most events. We will focus on these metrics: donations by industry, average donations per donor by industry, projected amount of donations per donor by industry.
Special Queries	Identifying sponsor participation via ordered parts: Sponsor_Participation <- SELECT COUNT(O.Placed_By_Company_Name) FROM Order O ORDER BY COUNT(Placed_By_Company_Name) ASC;	Project donations amount using an ARIMA or MA moving average model. We apply either 1) ARIMA time series or 1) MA moving average model (if we are unable to use time series) in R

Sample SQL Queries in Common for Queries 4 & 5 to Identify Current Donor & Sponsor Enthusiasm

Query	Join tables	Amount of donations by industry	Number of donors by industry	Identify average donations by donor(step 1)	Identify average donation quantity by donor (step 2)
Code	Joined <- SELECT D.DID, D.industry FROM Donors D LEFT JOIN Fund F ON D.DID=F.DID;	Donations <- SELECT SUM(j.amount) FROM Joined J GROUP BY j.industry ;	Count <- SELECT COUNT (J.DID) FROM Joined J GROUP BY J.industry;	Donations_average<- SELECT D.sum(donations) FROM Donations D LEFT JOIN Count C ON D.industry=C.industry;	Average_donations_by_industry<- SELECT sum(donations)\count(DID), industry FROM Donations_average GROUP BY industry ;